BENJAMIN YOUNG

Postdoctoral Researcher, Theoretical Computer Science Section, IT University of Copenhagen, Denmark
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EDUCATION

University of Wisconsin-Madison, Madison, WI, USA — September 2021 - June 2026
Ph.D. in Computer Science
Advisor: Prof. Jin-Yi Cai
Ph.D. Dissertation: *Indistinguishability: Counting, Constraints, and Contractions*

Case Western Reserve University, Cleveland, OH, USA — August 2017 - May 2021
B.S., M.S. in Computer Science, B.A. in Mathematics, *summa cum laude*
Advisor: Prof. Harold Connamacher
M.S. Thesis: *Totally Symmetric and Medial Quasigroups and their Applications*

HONORS AND AWARDS

Cisco Systems Distinguished Graduate Fellowship — 2025-2026
Department of Computer Sciences, UW-Madison

Best Student Paper, Track A, ICALP 2025 — 2025
For "The Converse of the Real Orthogonal Holant Theorem".

Graduate Student Instructor Award — 2023-2024
Department of Computer Sciences, UW-Madison

Graduate Research Travel Grant — 2023
Student Research Grants Competition, UW-Madison

Andrew R. Jennings Award for Excellence in Computer Science — 2021
Case Western Reserve University

Webster Godman Simon Award for Excellence in Mathematics — 2021
Case Western Reserve University

TEACHING EXPERIENCE

Grader, *Complexity Theory*, UW-Madison — Fall 2025
- Graded and provided written feedback on students' weekly lecture scribes and course projects. Managed course webpage and logistics.

Teaching Assistant, *An Introduction to the Theory of Computation*, UW-Madison — Spring 2025
- Held weekly office hours, graded and provided written feedback on exams and weekly homework. Managed course webpage and logistics. Prepared and gave one 75-minute guest lecture.

Grader/Guest Lecturer, *Graph Homomorphisms, #CSP, Tensor Networks*, UW-Madison — Fall 2024
- Graded and provided written feedback on students' weekly lecture scribes and course projects. Prepared and gave two 75-minute guest lectures.

Head Teaching Assistant, *Introduction to Algorithms*, UW-Madison — Fall 2022 - Spring 2023
- Led two weekly discussion sections of around 40 students each, and weekly study group with 5-10 students. Held 6 weekly office hours.
- Supervised undergraduate peer mentors.
- Wrote and graded quiz and exam questions, prepared homework assignments, managed course logistics.

Teaching Assistant, *Introduction to Algorithms*, UW-Madison Spring 2022 & Fall 2023 - Fall 2024

- Led one or two weekly discussion sections of around 40 students each, and weekly study group with 5-10 students. Held 6 weekly office hours. Wrote and graded quiz and exam questions.
- Supervised undergraduate peer mentors.

Teaching Assistant, *Introduction to Discrete Mathematics*, UW-Madison Fall 2021

- Led two weekly discussion sections of around 40 students each. Held 6 weekly office hours.
- Wrote and graded quiz and exam questions, prepared course materials.

Undergrad Teaching Assistant, *Theoretical Computer Science*, CWRU Spring 2020 & Spring 2021

- Held weekly office hours, graded and provided written feedback on weekly homework assignments.

PUBLICATIONS (Following convention, author order is alphabetical except for [6])

[1] **Vanishing Signatures, Orbit Closure, and the Converse of the Holant Theorem**.
Jin-Yi Cai and Ben Young
17th Innovations in Theoretical Computer Science Conference (ITCS 2026)

[2] **The Converse of the Real Orthogonal Holant Theorem**.
Ben Young
52nd International Colloquium on Automata, Languages, and Programming (ICALP 2025)
TheoretiCS 5.15 (Oct. 2026).

[3] **Quantum Algorithms for Discrete Log Require Precise Rotations**.
Jin-Yi Cai and Ben Young
ACM Transactions on Quantum Computing 6.3 (Sept. 2025)

[4] **Equality on all #CSP Instances Yields Constraint Function Isomorphism via Interpolation and Intertwiners**.
Ben Young
The Electronic Journal of Combinatorics 32 (Nov. 2025)

[5] **Planar #CSP Equality Corresponds to Quantum Isomorphism – A Holant Viewpoint**.
Jin-Yi Cai and Ben Young
50th International Colloquium on Automata, Languages, and Programming (ICALP 2023)
ACM Transactions on Computation Theory 16.3 (Sept. 2024).

[6] **The Number of Labeled n-ary Abelian Groups and Totally Symmetric Medial Quasigroups**.
Ben Young, Austin Hacker, and Harold Connamacher
Journal of Algebraic Combinatorics 57 (Jan. 2023).

RESEARCH EXPERIENCE

Postdoctoral Researcher, IT University of Copenhagen, Denmark July 2026 - present

Supervisor: Prof. Radu Curticapean
Study the mathematical properties and computational complexity of graph homomorphism counting and related counting problems.

Ph.D. Research, UW-Madison 2021 - 2026

Supervisor: Prof. Jin-Yi Cai
Investigated problems in and around the intersection of counting complexity, combinatorics, quantum computing, and quantum information theory. Discovered new mathematical properties of counting problems on graphs by expressing them as tensor network contractions.

Undergraduate and M.S. Research, Case Western Reserve University 2018 - 2021

Supervisor: Prof. Harold Connamacher

Wrote code to efficiently enumerate and search for structural patterns in totally symmetric and medial (TSM) quasigroups. Discovered a novel correspondence between TSM quasigroups, abelian groups, and their higher-order variants.

Research in Industrial Projects (RIPS) REU, University of California, Los Angeles Summer 2020

Led team of four students developing faster machine learning algorithms to solve elastic net regression on quantitative trading datasets with strict memory limitations.

TALKS

Guest Lecture, IT University of Copenhagen Sep. 15, 2026

90-minute lecture for Randomized Algorithms course.

CS Theory Seminar, Purdue University Feb. 19, 2026

"Planar #CSP Equality Corresponds to Quantum Isomorphism – A Holant Viewpoint"

ITCS 2026, Milan, Italy (conference presentation) Jan. 29, 2026

"Vanishing Signatures, Orbit Closure, and the Converse of the Holant Theorem"

Algorithms Group Seminar, IT University of Copenhagen Oct. 6, 2025

"Counting Indistinguishability and the Converse of the Holant Theorem"

ICALP 2025, Aarhus, Denmark (conference presentation) Jul. 9, 2025

"The Converse of the Real Orthogonal Holant Theorem"

Theory of Computing Seminar, UW-Madison Feb. 13, 2025

"The Converse of the Real Orthogonal Holant Theorem and Symmetric Tensor Diagonalization"

Combinatorics Seminar, UW-Madison Feb. 19, 2024

"General and Planar #CSP Equality Corresponds to Classical and Quantum Isomorphism"

ICALP 2023, Paderborn, Germany (conference presentation) Jul. 11, 2023

"Planar #CSP Equality Corresponds to Quantum Isomorphism – A Holant Viewpoint"

SERVICE

Journal Reviewer – *The Electronic Journal of Combinatorics* 2026

Conference Reviewer – *ITCS 2027* 2026

Conference Reviewer – *SOSA 2027* 2026

Conference Reviewer – *SODA 2027* 2026

Mentor – UW-Madison Computer Science Graduate School Summer Mentoring Program 2025

Conference Reviewer – *ICALP 2025* 2025

Journal Reviewer – *Advances in Combinatorics* 2024

Tutor – Provost Scholars Program, Case Western Reserve University 2017-2020

Provided one-on-one assistance to low-income East Cleveland high school students with homework assignments and ACT/SAT prep.